

Thermodynamics

Lecture 2B:

- Energy Transfer, General Energy Analysis

Prof. Claudio Hail (he/him)

After this lecture

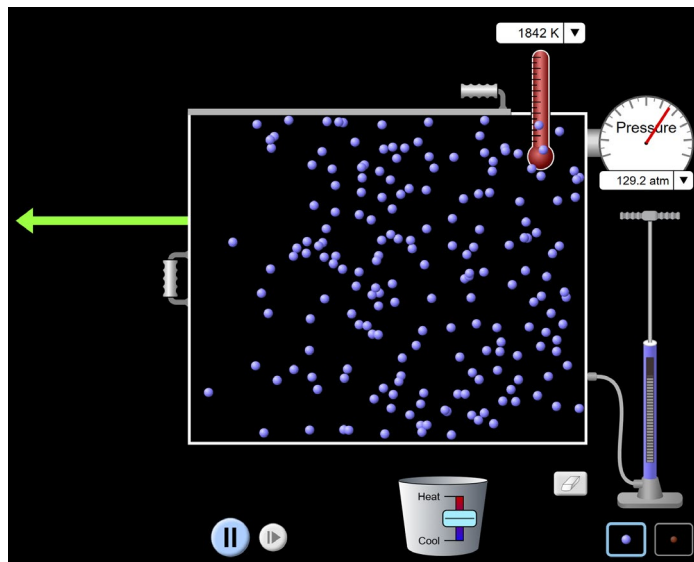
- Define the concept of work, including electrical and several forms of mechanical work.
- Understand the first law of thermodynamics, energy balances and mechanisms of energy transfer.
- Solving simple single-answer problems using the first law.

Learning objectives

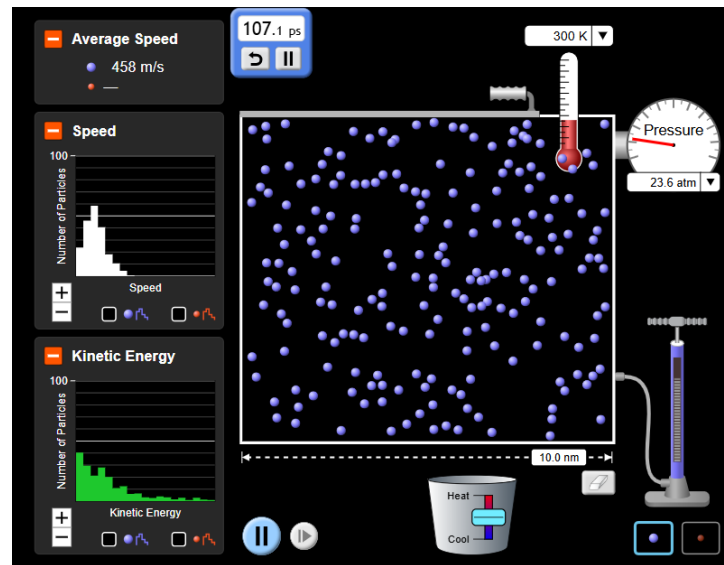
What you will be able to do after this course:

- I. *Knowledge:* Write the first and second laws of thermodynamics.**
- II. *Comprehension:* Describe the first and second laws in the student's own language.**
- III. *Application:* Solve simple single-answer problems using the first law.**
- IV. *Analysis:* Solve problems requiring both the first and second laws.*
- V. *Synthesis:* Develop schemes to improve the efficiency or performance of thermodynamic systems, e.g., compressors, turbines, heat pumps, etc.*

Recap demonstrations:



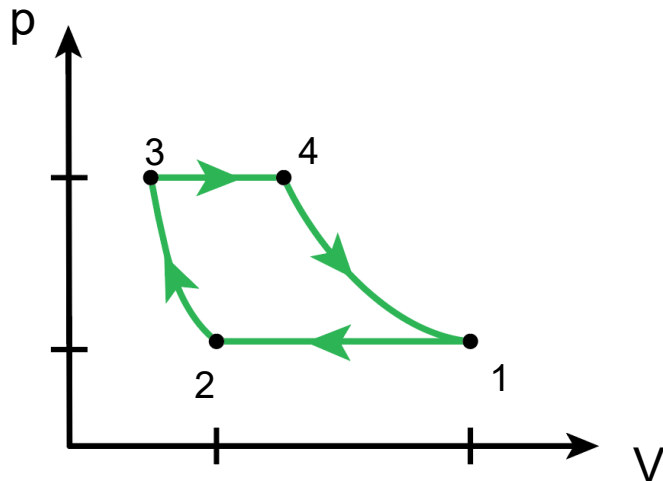
Non-quasistatic process



Measuring state variables

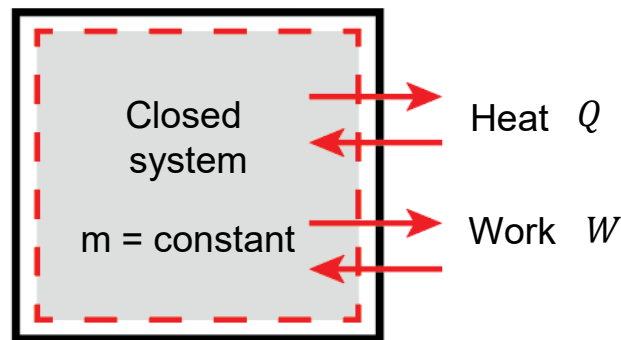
Recap Lecture 2A

- State properties ρ, T, p
- Internal energy U



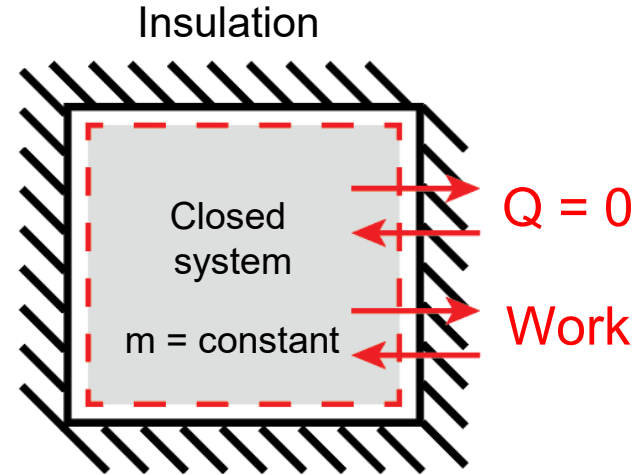
p-V diagram of a thermodynamic cycle

$$\Delta E = \Delta U + \Delta KE + \Delta PE + \dots$$



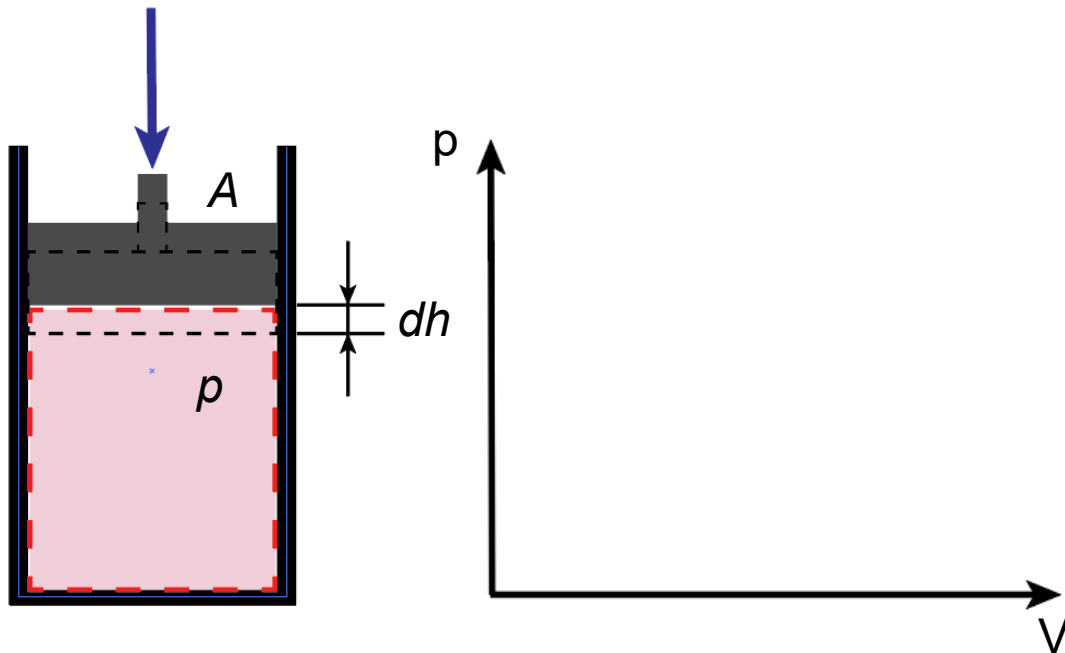
Adiabatic process

- A process where a system does not exchange heat with the surroundings.
- *Adiabatos*: not to be passed.



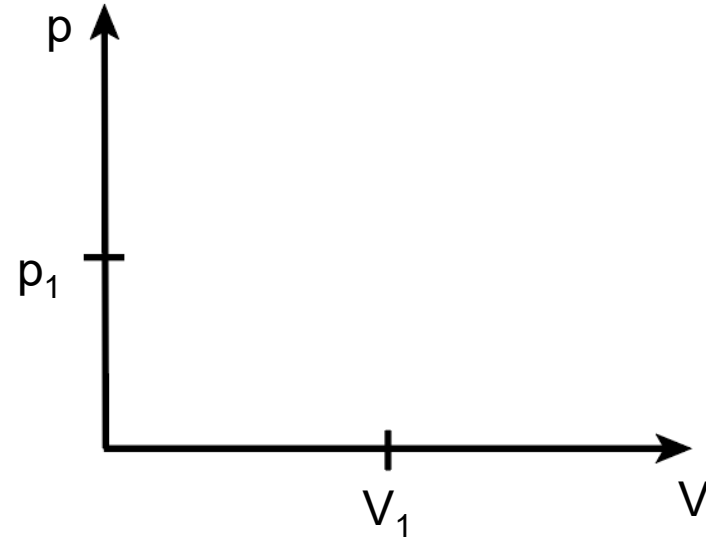
Energy transfer by work

- Work transfer on a closed system



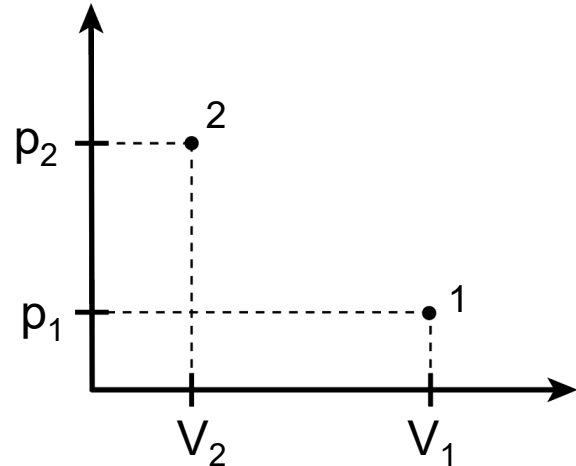
Thermodynamic state properties

- The properties of a state do not depend on the process that it was reached.
- State properties are point functions.



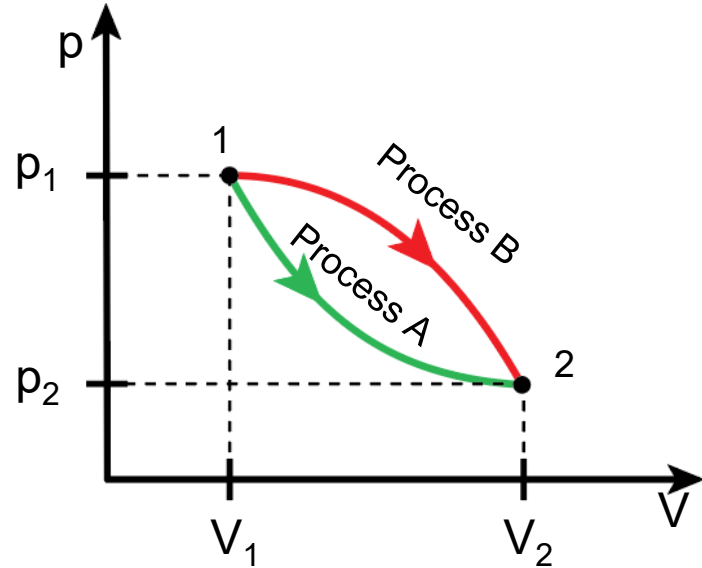
Energy transfer: Heat and work transfer

- Boundary phenomena: heat and work transfer occur at boundary of system.
- Are associated with a *process* not a state.
- Q and W depend on the *path* followed during the process, they are path functions



Energy Transfer: Heat and work transfer

- Q and W are path dependent.

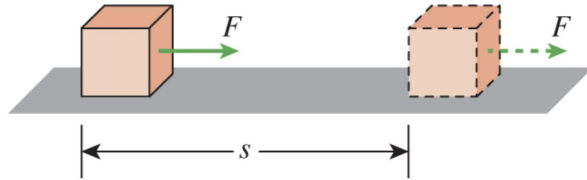


Types of mechanical work

- Linear mechanical work

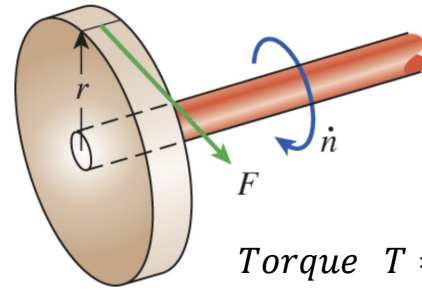
Force F

Distance s



- Shaft work

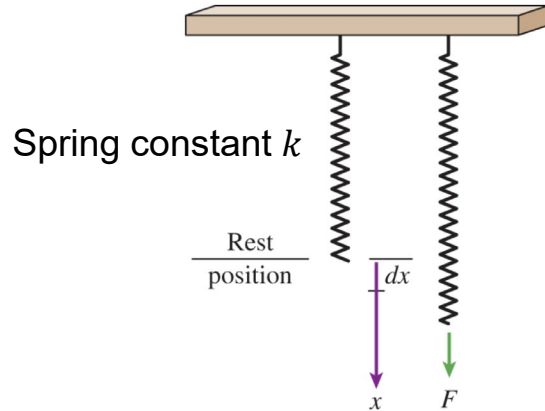
Rotational speed $\dot{n}$



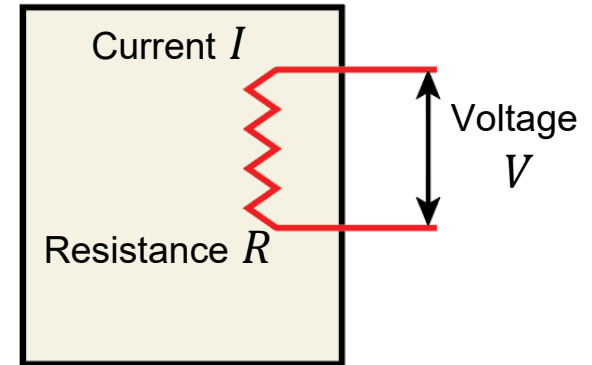
Torque $T = F r$

Types of work

- Spring work (mechanical)



- Electrical work



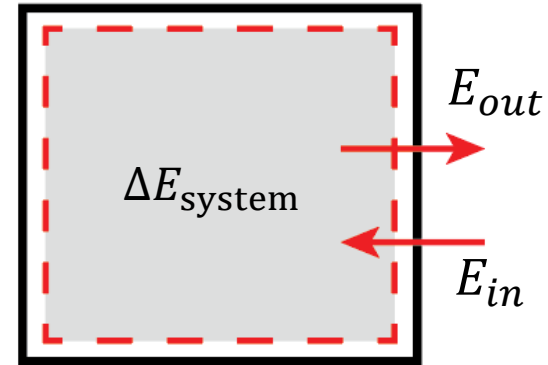
Example: Electric Water Kettle

Conservation of energy principle



Energy balance

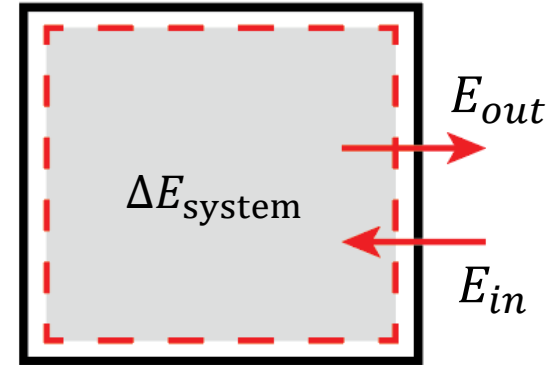
Conservation of energy principle



Energy change of a system

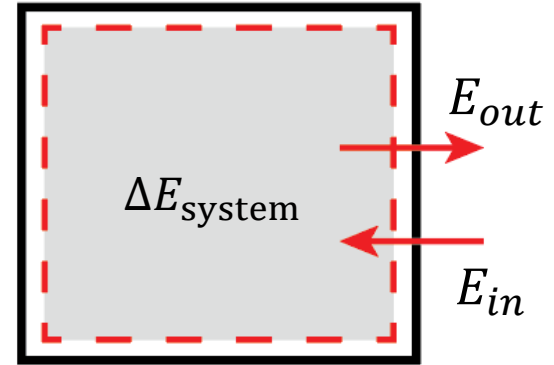
$$\Delta E_{\text{system}} =$$

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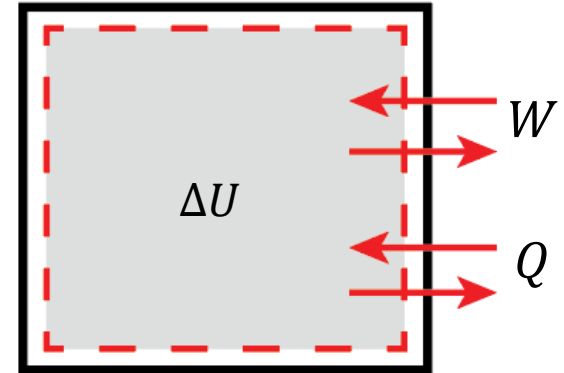
Energy transfer in/out of a closed system

- Account for energy transfer with surroundings.



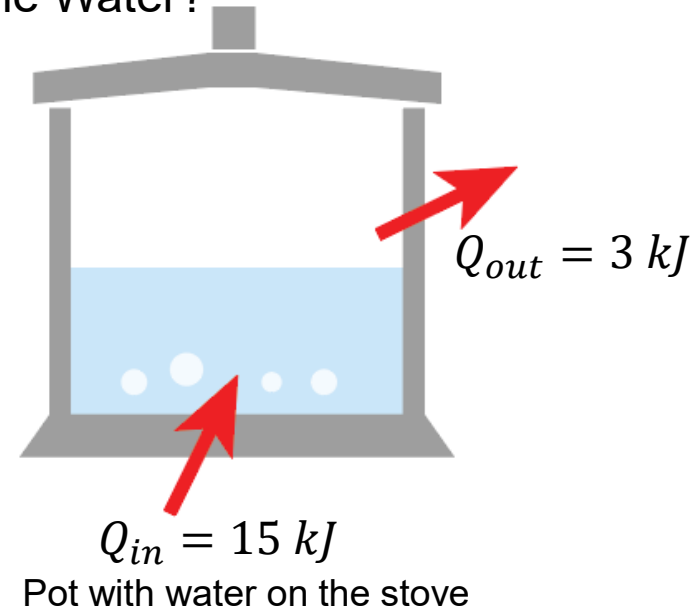
1st Law of thermodynamics: Closed system

$$E_{in} - E_{out} = \Delta E_{\text{system}}$$



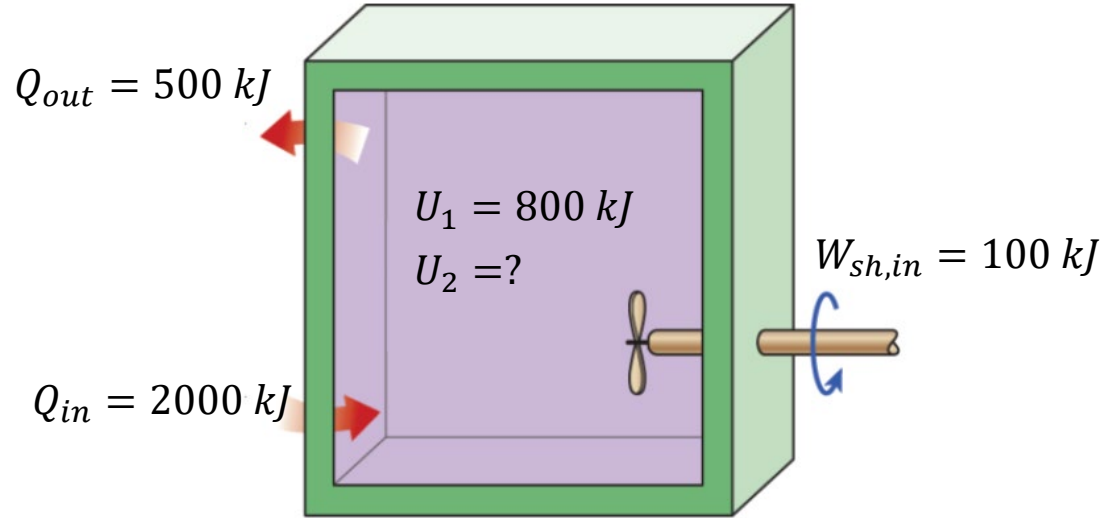
1st Law of thermodynamics: Example I

- Water is heated in a pot on a stove. Water experiences a heat transfer of $Q_{in} = 15 \text{ kJ}$ from the stove and loses $Q_{out} = 3 \text{ kJ}$ to the atmosphere via the pot. What is the change in internal energy of the Water?



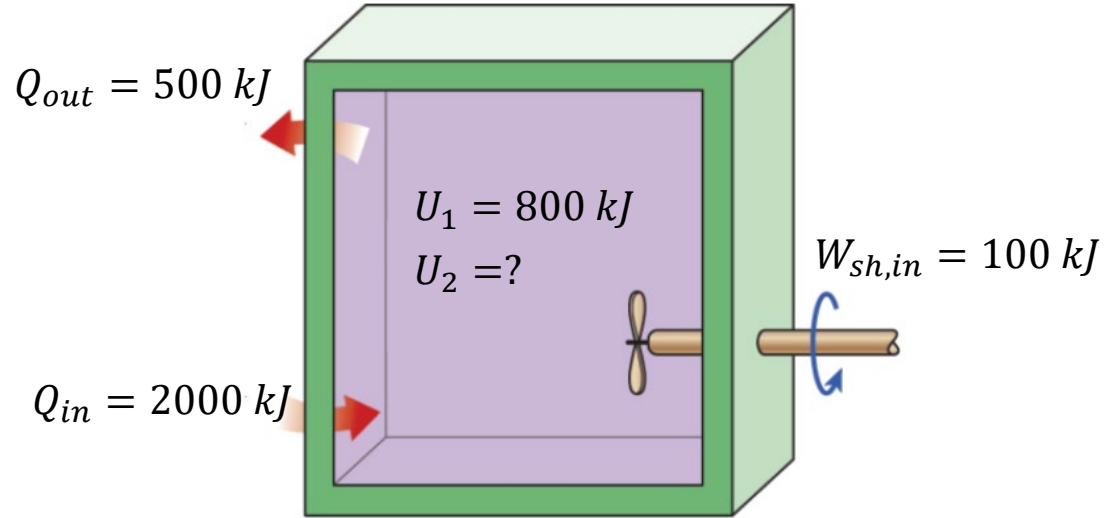
1st Law of thermodynamics: Example II

- Determine U_2 of the air inside the box.



1st Law of thermodynamics: Example II

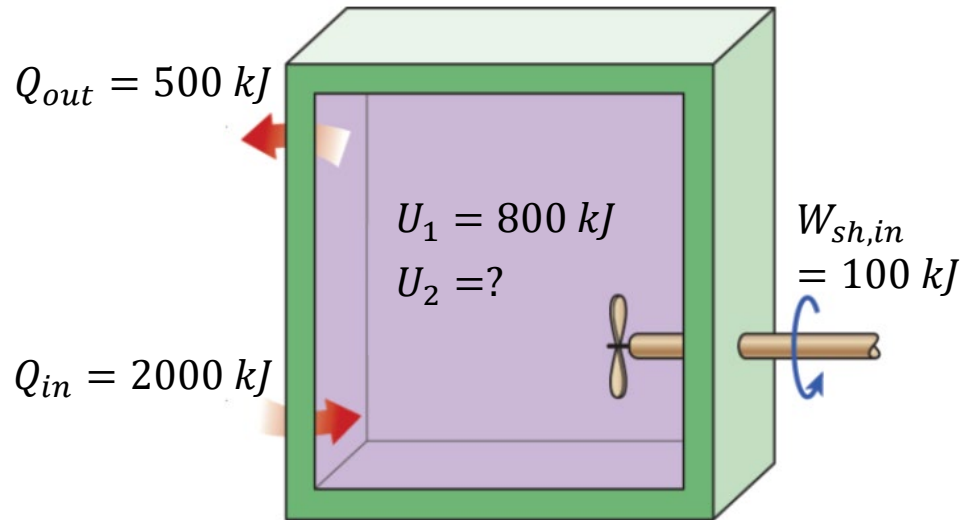
- Determine U_2 of the air inside the box.



Group Work: in groups of 2; solve the following problem.



Given:



Question:

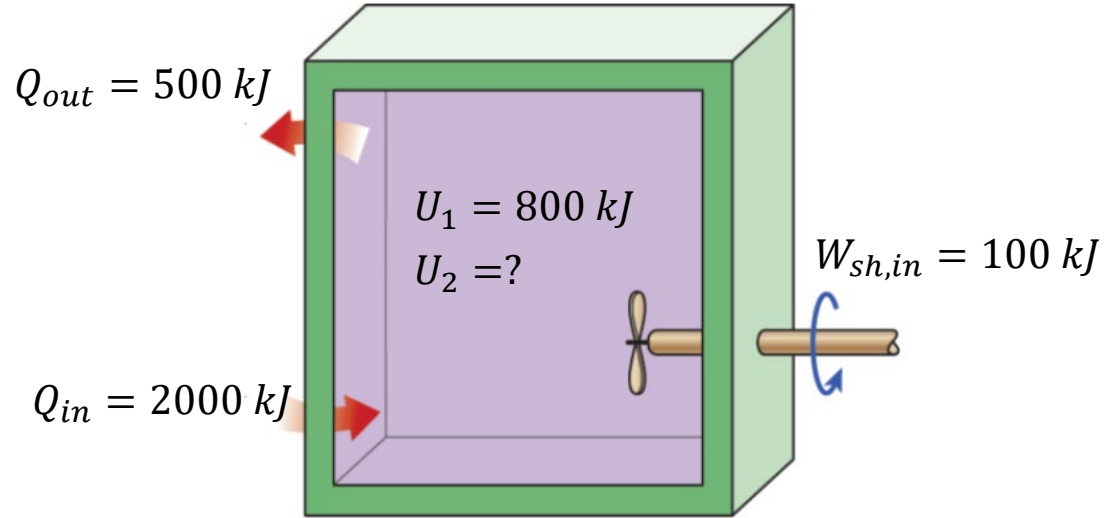
1. Determine U_2 of the air inside the box?

To Do:

- Define the thermodynamic system of interest.
- Write down an energy balance of the system.
- Go to pollev.com/claudiohail
- ... and answer question 1.

1st Law of thermodynamics: Example II

- Determine U_2 of the air inside the box.

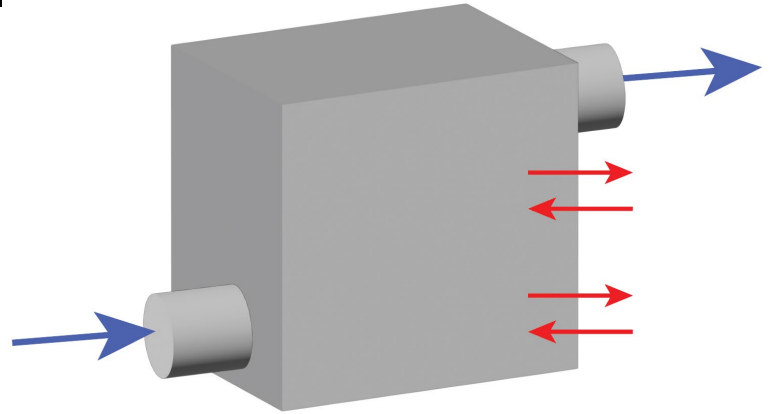


1st Law of thermodynamics: Open system

- Mass flows carry energy in/out.

$$E_{in} - E_{out} =$$

$$\Delta E_{\text{system}} =$$



Summary

- Discussed definitions of density, temperature and pressure.
- Understand the concept of energy and its different forms.
- Learn about the nature of internal energy.
- Concept of energy transfer by heat and work.

Next up

- Energy analysis of cycles
- Energy conversion efficiency